SM&I - Assignment 2

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April 9, 2026

Question 1

1. RStudio Console Output

```
> income <- c(25, 33, 22, 41, 18, 28, 32, 24, 53, 26)
> education <- c(11, 12, 11, 15, 8, 10, 11, 12, 17, 11)
>
> # Fit the linear regression model
> model_q1 <- lm(income ~ education)
> summary(model_q1)

Call:
lm(formula = income ~ education)

Residuals:
    Min      1Q  Median      3Q     Max
-6.9479 -1.9583  0.4219  3.0286  4.7917

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.9271     6.7802  -2.054 0.074038 .
education     3.7396     0.5631   6.641 0.000162 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.273 on 8 degrees of freedom
Multiple R-squared:  0.8465,    Adjusted R-squared:  0.8273
F-statistic: 44.11 on 1 and 8 DF,  p-value: 0.0001622

> new_edu <- data.frame(education = 16)
> predict(model_q1, new_edu, interval = "confidence", level = 0.90)
       fit     lwr     upr
1 45.90625 40.8413 50.9712
> predict(model_q1, new_edu, interval = "prediction", level = 0.90)
       fit      lwr      upr
1 45.90625 36.48281 55.32969
```

2. Discussion and Conclusion

a) LS Regression Line

Based on the R output, the estimated intercept (b_0) is -13.9271 and the estimated slope (b_1) is 3.7396. The Least Squares Regression Line for income on education level is:

$$\widehat{\text{Income}} = -13.9271 + 3.7396 \times \text{Education}$$

This implies that for every additional year of education, the individual's income is expected to increase by approximately $\$3,739.6$.

b) Standard Error of Estimate

The standard error of the estimate (S_e), which measures the typical distance between the observed values and the regression line, is found in the output as the "Residual standard error".

$$S_e = 4.273 \text{ (in \$1000s)}$$

c) Hypothesis Testing

We want to test if higher education leads to higher income.

- Null Hypothesis (H_0): $\beta_1 \leq 0$ (Education does not increase income).
- Alternative Hypothesis (H_a): $\beta_1 > 0$ (Higher education leads to higher income).

From the output, the two-tailed p-value for the slope is 0.000162. But since we are conducting a one-tailed test (checking if it is "higher"), the one-tailed p-value is $0.000162/2 = 0.000081$. It is much less than the significance level $\alpha = 0.05$, thus we strongly **reject the null hypothesis**. That is, the data provide sufficient evidence at the 5% significance level to conclude that a higher education level leads to a higher income.

d) Prediction Interval

For an individual with 16 years of education, the predicted income is 45.90625 ($\$45,906.25$). The question asks for the interval of "an individual", which in fact usually requires a **Prediction Interval** rather than a confidence interval for the mean.

- The 90% **Confidence Interval** (for the average income of all people with 16 years of education) is $[40.84, 50.97]$ (in \$1000s).
- The 90% **Prediction Interval** (for a single individual) is $[36.48, 55.33]$ (in \$1000s).

Thus, we can be 90% confident that a specific individual with 16 years of education will have an income between $\$36,482.81$ and $\$55,329.69$.

Question 2

1. RStudio Console Output

```
> batting <- c(0.254, 0.269, 0.255, 0.262, 0.254, 0.247, 0.264, 0.271,
   0.280, 0.256, 0.248, 0.255, 0.270, 0.257)
> winning <- c(0.414, 0.519, 0.500, 0.537, 0.352, 0.519, 0.506, 0.512,
   0.586, 0.438, 0.519, 0.512, 0.525, 0.562)
>
> # Fit the linear regression model
> model_q2 <- lm(winning ~ batting)
> summary(model_q2)

Call:
lm(formula = winning ~ batting)

Residuals:
      Min        1Q    Median        3Q       Max
-0.130908 -0.015261  0.005842  0.031416  0.070710

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.2268     0.4292  -0.528    0.607
batting       2.7941     1.6490   1.694    0.116

Residual standard error: 0.05669 on 12 degrees of freedom
Multiple R-squared:  0.1931,    Adjusted R-squared:  0.1258
F-statistic: 2.871 on 1 and 12 DF,  p-value: 0.116

> new_batting <- data.frame(batting = 0.275)
> predict(model_q2, new_batting, interval = "confidence", level = 0.90)
        fit       lwr      upr
1 0.5415841 0.4902453 0.592923
> predict(model_q2, new_batting, interval = "prediction", level = 0.90)
        fit       lwr     upr
1 0.5415841 0.4282583 0.65491
```

2. Discussion and Conclusion

a) LS Regression Line

Based on the R output, the estimated intercept (b_0) is -0.2268 and the estimated slope (b_1) is 2.7941. The Least Squares Regression Line for winning percentage on batting average is:

$$\widehat{\text{Winning}\%} = -0.2268 + 2.7941 \times \text{Batting\_Average}$$

b) Standard Error of Estimate

The standard error of the estimate (S_e) is found in the output as the "Residual standard error".

$$S_e = 0.05669$$

c) Hypothesis Testing

We want to test if a higher team batting average leads to a higher winning percentage.

- Null Hypothesis (H_0): $\beta_1 \leq 0$
- Alternative Hypothesis (H_a): $\beta_1 > 0$ (Higher batting average leads to higher winning percentage).

From the output, the two-tailed p-value for the slope is 0.116. Because our alternative hypothesis is directional (one-tailed test), the corresponding one-tailed p-value is calculated as $0.116/2 = 0.058$. It is greater than the significance level $\alpha = 0.05$, so we **fail to reject the null hypothesis**.
That is, the data do not provide sufficient evidence at the 5% significance level to conclude that a higher team batting average leads to a higher winning percentage.

d) Prediction Interval

For a team with a batting average of 0.275, the predicted winning percentage is 0.5416 (or 54.16%). Similar with 1d), although the question wants us to find the "confidence interval", since "a team" refers to an individual data point, the **Prediction Interval** might be a better statistical measure. Here we list both of them.

- The 90% **Confidence Interval** (for the true mean winning percentage of all teams with a 0.275 batting average) is $[0.4902, 0.5929]$.
- The 90% **Prediction Interval** (for one specific team with a 0.275 batting average) is $[0.4283, 0.6549]$.

Therefore, we can predict with 90% confidence that the winning percentage of a specific team whose batting average is 0.275 will fall between 42.83% and 65.49%.